

LAB 0 · SETUP CHEATSHEET**Lab Environment Setup**

Install & verify everything ONCE before Lab 1 — Docker · AWS CLI · Kubernetes (kind) · helper tools

What You Install (once)

Do this on your own laptop before the first lab. Everything runs locally — no cloud account, no credit card, no internet after the first downloads.

Tool	What it is for	Used in
Docker	Runs containers and the LocalStack cloud simulator	All labs
AWS CLI v2	Sends AWS commands to LocalStack	Labs 1, 3, 5
kind	Runs a local Kubernetes cluster inside Docker	Labs 1, 2, 4
kubectrl	Controls the Kubernetes cluster	Labs 1, 2, 4
OpenSSL	Encryption, keys, certificates	Lab 3
oathtool	Generates MFA / TOTP codes	Lab 4
Trivy	Scans containers for vulnerabilities (run via Docker — no install)	Lab 4

Security tip: Windows users: after installing Docker, do ALL lab commands inside Git Bash or WSL (Ubuntu). The labs use bash features (heredocs, sha256sum, single-quoting) that do not work in Command Prompt or PowerShell.

1. Install Docker

OS	How
Windows 10/11	Install Docker Desktop from docker.com. Choose the WSL 2 backend when prompted. Reboot.
macOS	Install Docker Desktop from docker.com (pick Apple Silicon or Intel to match your Mac).
Linux (Ubuntu)	Run: <code>curl -fsSL https://get.docker.com sh</code> then add yourself to the docker group (below).
<pre># Linux only: run docker without sudo (log out & back in after) sudo usermod -aG docker \$USER # Verify (all OS) – should print a version and a friendly message docker --version docker run --rm hello-world</pre>	

Caution: Docker needs hardware virtualization. If Docker won't start, enable virtualization (VT-x / AMD-V / SVM) in your laptop's BIOS/UEFI. On Windows, also enable WSL 2 and 'Virtual Machine Platform'.

2. Install AWS CLI v2

OS	How
Windows	Download and run the AWS CLI MSI installer from the AWS website (search 'install AWS CLI v2 Windows').
macOS	brew install awscli (or download the .pkg installer from AWS).
Linux	See the commands below.
<pre># Linux: install AWS CLI v2 curl 'https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip' -o awscliv2.zip unzip awscliv2.zip && sudo ./aws/install # Verify (all OS) aws --version # expect: aws-cli/2.x ...</pre>	

Note: You do NOT need a real AWS account. In the labs you point the AWS CLI at LocalStack with `--endpoint-url=http://localhost:4566`. See the separate 'AWS CLI Cheatsheet'.

3. Install kind & kubectl

OS	kind	kubectl
Windows	choco install kind	choco install kubernetes-cli
macOS	brew install kind	brew install kubectl
Linux	see commands below	sudo snap install kubectl --classic
<pre># Linux: install kind (Kubernetes-in-Docker) curl -Lo ./kind https://kind.sigs.k8s.io/dl/v0.23.0/kind-linux-amd64 chmod +x ./kind && sudo mv ./kind /usr/local/bin/kind # Verify (all OS) kind --version kubectl version --client</pre>		

4. Helper Tools (OpenSSL, oathtool, Trivy)

Tool	How to install
OpenSSL	Already installed on macOS/Linux. On Windows it comes with Git Bash — no separate install.
oathtool	macOS: brew install oath-toolkit · Linux: sudo apt install oathtool · Windows: use WSL, or a phone authenticator app.
Trivy	No install needed — Lab 4 runs it via Docker: <code>docker run --rm aquasec/trivy image <name></code> .
<pre># Verify (where applicable) openssl version oathtool --version # Lab 4 only</pre>	

5. Start & Stop the Lab Environment

LocalStack (the local AWS)

```
# Start LocalStack (first run downloads the image)
docker run -d --name localstack -p 4566:4566 localstack/localstack

# Check it is healthy
curl http://localhost:4566/_localstack/health

# Stop / start again / remove
docker stop localstack
docker start localstack
docker rm -f localstack          # remove completely
```

Kubernetes cluster (kind)

```
# Create a cluster (name it per the lab)
kind create cluster --name ccse

# Check it is up
kubectl cluster-info --context kind-ccse
kubectl get nodes

# Delete the cluster when finished
kind delete cluster --name ccse
```

6. One-Time AWS CLI Configuration

LocalStack accepts any credentials. Set dummy values once so the CLI stops asking:

```
aws configure set aws_access_key_id test
aws configure set aws_secret_access_key test
aws configure set region us-east-1

# Save typing: put the endpoint in a variable each session
EP='--endpoint-url=http://localhost:4566'
aws $EP sts get-caller-identity      # proves the CLI is talking to LocalStack
```

Security tip: Optional shortcut: install 'awslocal' (pip install awscli-local) so you can type awslocal instead of aws \$EP. Both do the same thing.

7. Pre-Lab Verification Checklist

Tick each before Lab 1. If any fails, see Troubleshooting below.

- ☐ docker --version prints a version, and docker run hello-world works.
- ☐ aws --version prints aws-cli/2.x.

- ☐ kind --version and kubectl version --client both work.
- ☐ LocalStack starts and curl .../health responds.
- ☐ aws \$EP sts get-caller-identity returns an identity.
- ☐ kind create cluster works and kubectl get nodes shows a node.
- ☐ (Windows) You are working inside Git Bash or WSL.

8. Troubleshooting

Symptom	Fix
'Cannot connect to the Docker daemon'	Docker Desktop is not running — start it. On Linux, run the usermod command and re-login.
Docker won't start / very slow	Enable virtualization in BIOS. On Windows enable WSL 2 + Virtual Machine Platform.
Port 4566 already in use	A LocalStack is already running: docker rm -f localstack, then start again.
aws: 'Could not connect to the endpoint URL'	LocalStack is not running, or you forgot --endpoint-url / \$EP. Start LocalStack and retry.
'aws: command not found' / 'kubectl not found'	The tool is not installed or not on PATH. Re-run the install step; open a new terminal.
heredoc / sha256sum errors on Windows	You are in PowerShell/CMD. Switch to Git Bash or WSL.
kind create cluster fails	Docker not running, or low memory. Ensure Docker has ≥ 4 GB and is started.
Image download is slow in the lab	Ask the instructor to pre-pull images, or run the docker run/pull commands before class on Wi-Fi.
MFA/TOTP code always fails (Lab 4)	Your system clock is out of sync — enable automatic time, then retry.
NetworkPolicy not blocking (Lab 2)	The cluster needs Calico (the Lab 2 setup installs it). Wait until calico-node is Ready.

9. Handy One-Liners

```
# Start everything for a session
docker start localstack 2>/dev/null || docker run -d --name localstack -p 4566:4566
localstack/localstack
EP='--endpoint-url=http://localhost:4566'

# See what is running
docker ps
kind get clusters

# Clean up everything (frees disk)
docker rm -f localstack
kind delete clusters --all
```

```
docker system prune -f          # removes unused images/containers
```

Note: Removing containers/clusters is safe — you recreate them from the lab commands next time. Your lab report files (screenshots, outputs) are saved on your laptop, not in the containers.